



Dayananda Sagar University

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IntentCloud: Intent-Aware Cognitive Cloud Memory System

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Abstract

- Traditional file storage systems rely on filenames, folders, and keyword-based search, making document retrieval difficult when users cannot remember exact file details.
- Existing AI-powered retrieval solutions often depend on cloud services, leading to privacy concerns, recurring costs, and limited offline accessibility.
- IntentCloud is a self-hosted, intent-aware cognitive cloud memory system that enables users to retrieve documents using natural language queries instead of exact keywords.
- The proposed system combines local AI-based intent understanding, semantic embeddings, and vector similarity search to improve the relevance and accuracy of file retrieval.
- A Raspberry Pi provides secure edge storage, while Cloudflare Tunnel enables protected remote access without exposing the local network.
- The proposed architecture aims to deliver privacy-preserving, cost-effective, and efficient semantic file retrieval on consumer-grade hardware.



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Introduction

- Existing storage systems organize files using filenames and folders.
- Users often forget filenames after several months.
- Searching by keyword may return many unrelated files.
- Semantic search understands the context of a query.
- IntentCloud works like a digital memory assistant.
- It retrieves documents based on user intent instead of exact words.
- The system is designed to be private, secure, and self-hosted.
- Existing semantic retrieval solutions do not simultaneously provide intent-aware retrieval, local AI processing, secure storage, and remote accessibility.



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Literature Survey



Sl. NO.	Paper Title	Methodology & Accuracy	Pro's	Con's
1	Semantic Search in a Document Management System Using Retrieval-Augmented Generation (Krishnan & Subramanian, 2025)[1]	Uses RAG to understand the user's question and search documents by meaning instead of keywords. Gives accurate results but depends on cloud AI.	Finds relevant documents faster than normal keyword search.	Needs paid cloud APIs, so user data is not completely private.
2	Constructing ChatGPT for PDF Files with LangChain (Jacob et al., 2024)[2]	Splits PDF into small parts, converts them into embeddings, and uses ChatGPT to answer user questions. Works well for document search.	Easy to search and understand PDF content using natural language.	Breaking documents into small chunks may miss the complete context of the document.
3	Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Retrieval (Sawarkar et al., 2024)[3]	Combines keyword search and semantic search together to improve retrieval accuracy.	Better search accuracy because it uses two search methods together.	Needs a large amount of memory and powerful hardware for big datasets.
4	Efficient RAG with Intent-Aware Retrieval and Semantic-Preserving Chunking (Puspitasari et al., 2026)[4]	Understands the user's intent before searching and stores document chunks carefully to improve retrieval quality.	Gives more relevant search results by understanding user intention.	Works well only when information is inside nearby chunks; struggles with information spread across a document.
5	Research on an Intelligent File Query System Integrating Natural Language Understanding and Information Retrieval (Nan et al., 2026)[5]	Uses Natural Language Understanding (NLU) and semantic retrieval to find files from user queries. Provides very high search accuracy.	Very accurate file retrieval using AI and semantic understanding.	Requires expensive GPUs and high-end hardware, making it difficult for normal users.



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Problem Statement & Objectives

Problem Statement

- Students, researchers, and professionals often struggle to locate documents because they cannot remember exact filenames, folder locations, or keywords.
- As the number of stored files increases, traditional keyword-based search becomes inefficient, leading to longer search times and reduced productivity.
- Existing semantic retrieval systems often depend on cloud services, resulting in privacy concerns, recurring subscription costs, and limited offline accessibility.
- Current solutions do not provide a complete self-hosted, intent-aware document retrieval system that combines privacy, secure storage, and remote accessibility on consumer-grade hardware.
- Therefore, there is a need for a privacy-preserving, self-hosted, intent-aware semantic file retrieval system that enables users to search documents using natural language while ensuring secure and efficient access.



Problem Statement & Objectives

Objectives

- Design and develop an AI-powered semantic document retrieval system to improve file retrieval accuracy compared with traditional keyword-based search.
- Develop a privacy-preserving document management system through secure local AI-based processing and storage.
- Provide secure and efficient remote access to documents through a self-hosted architecture.
- Develop a lightweight self-hosted solution that operates efficiently on consumer-grade hardware.



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Proposed Methodology

Major Features of the Proposed System

- Natural Language Search – Retrieve files using conversational queries.
- Intent-Aware Retrieval – Understands user intent beyond keywords.
- Semantic Search – Finds documents based on meaning using AI embeddings.
- Local AI Processing – Performs inference offline using Phi-3 Mini.
- Secure Edge Storage – Stores files securely on Raspberry Pi.
- Remote Access – Provides secure access through Cloudflare Tunnel.
- Privacy-Preserving – Eliminates dependence on third-party cloud APIs.

Proposed Methodology

Proposed IntentCloud System Architecture

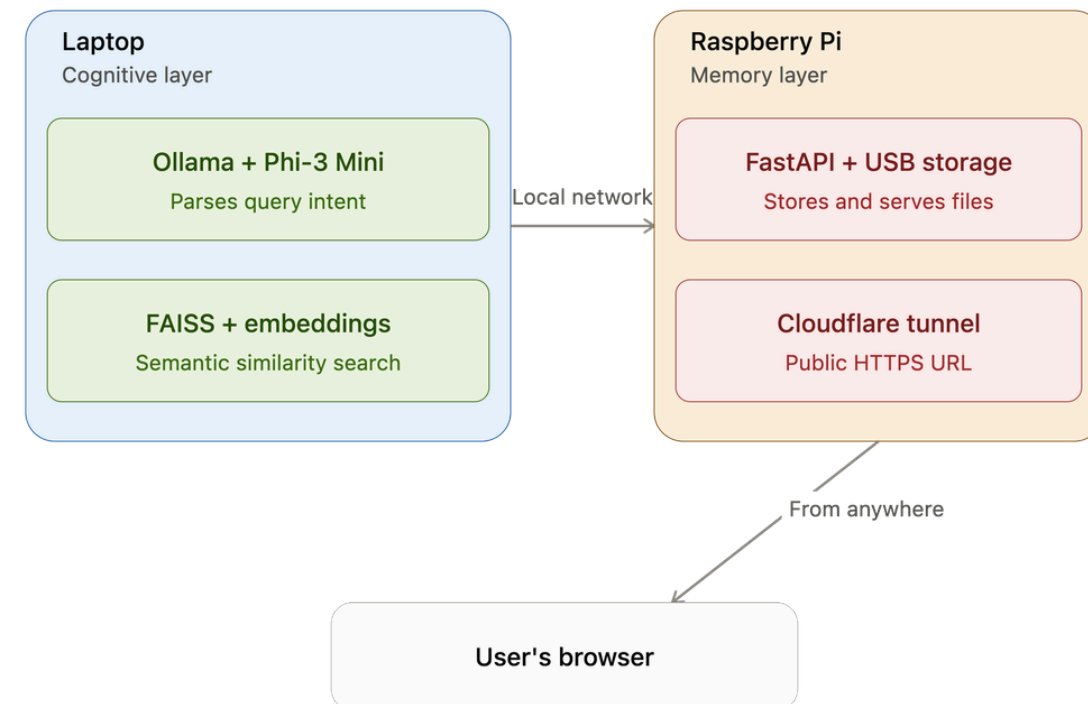


Figure 1. Proposed IntentCloud architecture showing the interaction between the User Interface, Cognitive Layer, Memory Layer, and Cloud Access

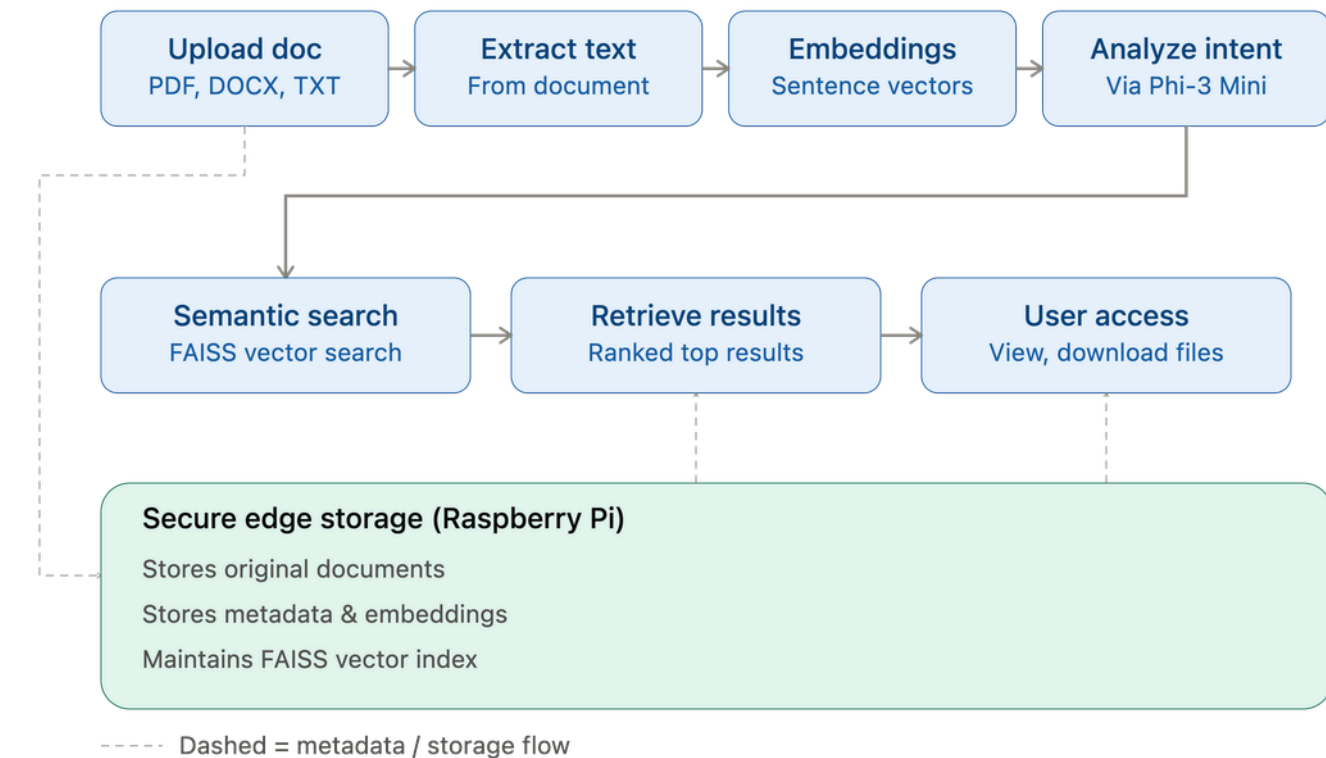


Figure 2 .Workflow of IntentCloud for intent-aware semantic file retrieval.



References

- [1]Krishnan, S., & Subramanian, N., "Semantic Search in a Document Management System Using Retrieval Augmented Generation," IEEE/ACIS SNPD, 2025.
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- [3]Sawarkar, K., Mangal, A., & Solanki, S. R., "Blended RAG: Improving RAG Accuracy with Semantic Search and Hybrid Query-Based Retrievers," arXiv, 2024.
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- [5]Nan, Y., et al., "Research on an Intelligent File Query System Model Integrating Natural Language Understanding and Information Retrieval," IEEE International Conference on Power, Electronics and Green Energy (ICPEGE), 2026.



Thank You